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EXAMINER	
RIDDER, CLAYTON PAUL	

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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/501,015	Applicant(s) Chi, Hsiang-Feng	
	Examiner CLAYTON P RIDDER	Art Unit 3646	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 12/22/2025.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-19 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-19 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Response to Arguments

1. Applicant's arguments filled 12/22/2025, have been considered but are moot because the new ground of rejection does not solely rely on any reference applied in the prior rejection of record for any teaching or matter specifically challenged in the argument.

Claim Rejections - 35 USC § 103

2. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

3. The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

4. **Claims 1, 4-9, 11-12, 15-19** are rejected under 35 U.S.C. 103 as being unpatentable over **LAGHEZZA(CN111880169A)** in view of **Matsugatani(US6040796A).**

Regarding claim 1, LAGHEZZA discloses

A radar apparatus, comprising: a frequency synthesizer, configured to generate a carrier signal **("PLL111 generates ramp FMCW radar signal" [0049])**, wherein the carrier signal has a frequency parameter **("frequency modified carrier signal is generated" [0056])**, the frequency parameter of the carrier signal

changes with time within a sweep period of the carrier signal **("the transmitting-side circuit generates a radar waveform sequence by sweeping a carrier signal over a frequency range" [0045])**, and the carrier signal comprises a frequency raising section and a frequency decreasing section **("an FMCW signal that may be transmitted as a ramped radar chirp" [0040])**; a signal generator, configured to generate a shaping signal **(" the PSK input can be provided by the reduced angle PSK (Mr-PSK) module 113." [0049]),** [...] and a transmitting front-end circuit, coupled to the frequency synthesizer and the signal generator and configured to generate a transmitting signal according to the frequency parameter of the carrier signal and the [...] shaping signal **(FIG.1, Part 115)**

LAGHEZZA does not explicitly disclose wherein, a maximum value of the maximum amplitude parameter in a first section is greater than a maximum value of the maximum amplitude parameter in a second section. Matsugatani teaches in the same field of endeavor of carrier signal modulation. Matsugatani discloses wherein, the shaping signal has a maximum amplitude parameter **("local modulation signal Mb is used for an amplitude modulation of the branch signal " [Col.27, ll.11-12])** the shaping signal comprises a first section and a second section, the first section and the second section have different amplitudes, and the second section corresponds to a turn-around section between the frequency raising section and the frequency decreasing section **("it is also preferable that the modulation signal oscillator 54 produces a rectangular wave including many harmonics having frequencies n.times.Fp (n.gtoreq.2)" [Col.30, ll.9-12])**, and a maximum value of the maximum amplitude parameter in the first section is greater than a maximum value of the maximum amplitude parameter in the second section **("the cyclic modulation signal Mb can be a rectangular or triangular wave when it can causes a cyclic variation in the frequency of the transmitting signal Ss." [col.14, ll.40-43] The Examine notes that the rectangular modulation signal necessarily has two sections, one corresponding to the constant peak of the wave, and the second corresponding to the turn around**

section of the rectangular wave. As the second section corresponds to the "turn around" of the wave, its maximum amplitude is lower than the first section).

Matsugatani teaches in the same field of endeavor of carrier signal modulation. It would have been obvious to one of ordinary skill in the art prior to the effective filing date of the claimed invention to modify **LAGHEZZA** with the teachings of **Matsugatani** to incorporate the features of, a maximum value of the maximum amplitude parameter in a first section of a shaping signal being greater than a maximum value of the maximum amplitude parameter in a second section of a shaping signal so as to gain the advantage of improving detection ability [Col.19, Par.5, Matsugatani] . Also, since it has been held that if a technique has been used to improve one device, and a person of ordinary skill in the art would recognize that it would improve similar devices in the same way, using the technique is obvious unless its actual application is beyond his or her skill (MPEP 2143).

Regarding claim 4, LAGHEZZA as modified by Matsugatani discloses all the limitations of claim 1. LAGHEZZA does not explicitly disclose wherein, a maximum value of the maximum amplitude parameter in a first section is greater than a maximum value of the maximum amplitude parameter in a second section. Matsugatani teaches in the same field of endeavor of carrier signal modulation. Matsugatani discloses wherein, an amplitude of the second section is smaller than the first section **("it is also preferable that the modulation signal oscillator 54 produces a rectangular wave including many harmonics having frequencies n.times.Fp (n.gtoreq.2)" [Col.30, ll.9-12])**

Matsugatani teaches in the same field of endeavor of carrier signal modulation. It would have been obvious to one of ordinary skill in the art prior to the effective filing date of the claimed invention to modify **LAGHEZZA** with the teachings of **Matsugatani** to incorporate the features of an amplitude of the second section being smaller than the first section so as to gain the advantage of improving detection ability [Col.19, Par.5, Matsugatani] . Also, since it has been held that if a technique has been

used to improve one device, and a person of ordinary skill in the art would recognize that it would improve similar devices in the same way, using the technique is obvious unless its actual application is beyond his or her skill (MPEP 2143).

Regarding claim 5, LAGHEZZA as modified by Matsugatani discloses all the limitations of claim 1. LAGHEZZA does not explicitly disclose wherein, a maximum value of the maximum amplitude parameter in a first section is greater than a maximum value of the maximum amplitude parameter in a second section. Matsugatani teaches in the same field of endeavor of carrier signal modulation. Matsugatani discloses wherein, a waveform of the second section is defined by a sin-squared function, a linear function, or a rectangular function **("it is also preferable that the modulation signal oscillator 54 produces a rectangular wave including many harmonics having frequencies n.times.Fp (n.gtoreq.2)" [Col.30, ll.9-12])**

Matsugatani teaches in the same field of endeavor of carrier signal modulation. It would have been obvious to one of ordinary skill in the art prior to the effective filing date of the claimed invention to modify **LAGHEZZA** with the teachings of **Matsugatani** to incorporate the features of a rectangular function so as to gain the advantage of improving detection ability [Col.19, Par.5, Matsugatani] . Also, since it has been held that if a technique has been used to improve one device, and a person of ordinary skill in the art would recognize that it would improve similar devices in the same way, using the technique is obvious unless its actual application is beyond his or her skill (MPEP 2143).

Regarding claim 6, LAGHEZZA as modified by Matsugatani discloses all the limitations of claim 1. LAGHEZZA does not explicitly disclose wherein an amplitude is the same during the first section at a time domain. Matsugatani teaches in the same field of endeavor of carrier signal modulation. Matsugatani discloses wherein, an amplitude is the same during the first section at a time domain **("it is**

also preferable that the modulation signal oscillator 54 produces a rectangular wave including many harmonics having frequencies n.times.Fp (n.gtoreq.2)" [Col.30, ll.9-12])

Matsugatani teaches in the same field of endeavor of carrier signal modulation. It would have been obvious to one of ordinary skill in the art prior to the effective filing date of the claimed invention to modify **LAGHEZZA** with the teachings of **Matsugatani** to incorporate the features of an amplitude being the same during the first section at a time domain so as to gain the advantage of improving detection ability [Col.19, Par.5, Matsugatani] . Also, since it has been held that if a technique has been used to improve one device, and a person of ordinary skill in the art would recognize that it would improve similar devices in the same way, using the technique is obvious unless its actual application is beyond his or her skill (MPEP 2143).

Regarding claim 7, LAGHEZZA as modified by Matsugatani discloses all the limitations of claim 1.

LAGHEZZA discloses wherein, the transmitting front-end circuit comprises: a first mixer, coupled to the signal generator and the frequency synthesizer **(FIG.1, Part.112)** and configured to mix the carrier signal and the shaping signal to generate a mixed signa **("The mixer 112 uses the PSK signal from the Mr-PSK module 113 to mix the ramp FMCW"[0050])**l; and a first amplifier, coupled to a first antenna and configured to amplify the mixed signal to generate the transmitting signal **("power amplifier 114" [0049])**, and the radar apparatus further comprising: the first antenna, coupled to the first amplifier and configured to emit the transmitting signal **("The power amplifier 114 amplifies the output of the mixer 112 to be transmitted via the antenna 115" [0049]).**

Regarding claim 8, LAGHEZZA as modified by Matsugatani discloses all the limitations of claim 1.

LAGHEZZA discloses, a controller, coupled to the signal generator and the frequency synthesizer and configured to synchronize the second section to the turn-around section between the frequency raising

section and the frequency decreasing section **("The transmitting-side circuit 110 and/or the receiving-side circuit 120 may be synchronized using the timing block 130 ." [0050])**

Regarding claim 9, LAGHEZZA as modified by Matsugatani discloses all the limitations of claim 1. LAGHEZZA discloses wherein, the signal generator comprises: a wave generator, configured to generate an initial shaping signal **("the transmit-side circuit includes an oscillator" [0045])**; and a digital to analog converter, coupled to the wave generator and configured to convert the initial shaping signal into the shaping signal **("the transmit-side circuitry includes a digital signal processing circuit for processing a digitized version of the communication data and a mixer for performing frequency up-conversion." [0042]).**

Regarding claim 11, LAGHEZZA discloses

A signal processing circuit, comprising: a frequency synthesizer, configured to generate a carrier signal **("PLL111 generates ramp FMCW radar signal" [0049])**, wherein the carrier signal has a frequency parameter **("frequency modified carrier signal is generated" [0056])** the frequency parameter of the carrier signal changes with time within a sweep period of the carrier signal **("the transmitting-side circuit generates a radar waveform sequence by sweeping a carrier signal over a frequency range" [0045])**, and the carrier signal comprises a frequency raising section and a frequency decreasing section **("an FMCW signal that may be transmitted as a ramped radar chirp" [0040])**; and a signal generator, configured to generate a shaping signal **(" the PSK input can be provided by the reduced angle PSK (Mr-PSK) module 113." [0049]).**

LAGHEZZA does not explicitly disclose wherein, a maximum value of the maximum amplitude parameter in a first section is greater than a maximum value of the maximum amplitude parameter in a second section. Matsugatani teaches in the same field of endeavor of carrier signal modulation.

Matsugatani discloses wherein, the shaping signal has a maximum amplitude parameter **("local modulation signal Mb is used for an amplitude modulation of the branch signal " [Col.27, ll.11-12])** the shaping signal comprises a first section and a second section, the first section and the second section have different amplitudes, and the second section corresponds to a turn-around section between the frequency raising section and the frequency decreasing section **("it is also preferable that the modulation signal oscillator 54 produces a rectangular wave including many harmonics having frequencies n.times.Fp (n.gtoreq.2)" [Col.30, ll.9-12])**, and a maximum value of the maximum amplitude parameter in the first section is greater than a maximum value of the maximum amplitude parameter in the second section **("the cyclic modulation signal Mb can be a rectangular or triangular wave when it can causes a cyclic variation in the frequency of the transmitting signal Ss." [col.14, ll.40-43] The Examine notes that the rectangular modulation signal necessarily has two sections, one corresponding to the constant peak of the wave, and the second corresponding to the turn around section of the rectangular wave. As the second section corresponds to the "turn around" of the wave, its maximum amplitude is lower than the first section).**

Matsugatani teaches in the same field of endeavor of carrier signal modulation. It would have been obvious to one of ordinary skill in the art prior to the effective filing date of the claimed invention to modify **LAGHEZZA** with the teachings of **Matsugatani** to incorporate the features of, a maximum value of the maximum amplitude parameter in a first section of a shaping signal being greater than a maximum value of the maximum amplitude parameter in a second section of a shaping signal so as to gain the advantage of improving detection ability [Col.19, Par.5, Matsugatani] . Also, since it has been held that if a technique has been used to improve one device, and a person of ordinary skill in the art would recognize that it would improve similar devices in the same way, using the technique is obvious unless its actual application is beyond his or her skill (MPEP 2143).

Regarding claim 12, LAGHEZZA discloses

A signal processing method for a radar apparatus, comprising: generating a carrier signal **("PLL111 generates ramp FMCW radar signal" [0049])**, wherein the carrier signal has a frequency parameter **("frequency modified carrier signal is generated" [0056])**, the frequency parameter of the carrier signal changes with time within a sweep period of the carrier signal **("the transmitting-side circuit generates a radar waveform sequence by sweeping a carrier signal over a frequency range" [0045])**, and the carrier signal comprises a frequency raising section and a frequency decreasing section **("an FMCW signal that may be transmitted as a ramped radar chirp" [0040])**; generating a shaping signal **(" the PSK input can be provided by the reduced angle PSK (Mr-PSK) module 113." [0049])**, and generating a transmitting signal according to the frequency parameter of the carrier signal and [...] the shaping signal, wherein the transmitting signal is configured to allow the radar apparatus to emit **(FIG.1, Part 115).**

LAGHEZZA does not explicitly disclose wherein, a maximum value of the maximum amplitude parameter in a first section is greater than a maximum value of the maximum amplitude parameter in a second section. Matsugatani teaches in the same field of endeavor of carrier signal modulation. Matsugatani discloses wherein, the shaping signal has a maximum amplitude parameter **("local modulation signal Mb is used for an amplitude modulation of the branch signal " [Col.27, ll.11-12])** the shaping signal comprises a first section and a second section, the first section and the second section have different amplitudes, and the second section corresponds to a turn-around section between the frequency raising section and the frequency decreasing section **("it is also preferable that the modulation signal oscillator 54 produces a rectangular wave including many harmonics having frequencies n.times.Fp (n.gtoreq.2)" [Col.30, ll.9-12])**, and a maximum value of the maximum amplitude parameter in the first section is greater than a maximum value of the maximum amplitude parameter in the second section **("the cyclic modulation signal Mb can be a rectangular or triangular wave when it can causes a cyclic variation in the frequency of the transmitting signal Ss." [col.14, ll.40-**

43] The Examiner notes that the rectangular modulation signal necessarily has two sections, one corresponding to the constant peak of the wave, and the second corresponding to the turn around section of the rectangular wave. As the second section corresponds to the "turn around" of the wave, its maximum amplitude is lower than the first section).

Matsugatani teaches in the same field of endeavor of carrier signal modulation. It would have been obvious to one of ordinary skill in the art prior to the effective filing date of the claimed invention to modify **LAGHEZZA** with the teachings of **Matsugatani** to incorporate the features of, a maximum value of the maximum amplitude parameter in a first section of a shaping signal being greater than a maximum value of the maximum amplitude parameter in a second section of a shaping signal so as to gain the advantage of improving detection ability [Col.19, Par.5, Matsugatani] . Also, since it has been held that if a technique has been used to improve one device, and a person of ordinary skill in the art would recognize that it would improve similar devices in the same way, using the technique is obvious unless its actual application is beyond his or her skill (MPEP 2143).

Regarding claim 15, LAGHEZZA as modified by Matsugatani discloses all the limitations of claim 12. LAGHEZZA does not explicitly disclose wherein, a maximum value of the maximum amplitude parameter in a first section is greater than a maximum value of the maximum amplitude parameter in a second section. Matsugatani teaches in the same field of endeavor of carrier signal modulation. Matsugatani discloses wherein, an amplitude of the second section is smaller than the first section **("it is also preferable that the modulation signal oscillator 54 produces a rectangular wave including many harmonics having frequencies n.times.Fp (n.gtoreq.2)" [Col.30, ll.9-12])**

Matsugatani teaches in the same field of endeavor of carrier signal modulation. It would have been obvious to one of ordinary skill in the art prior to the effective filing date of the claimed invention to modify **LAGHEZZA** with the teachings of **Matsugatani** to incorporate the features of an amplitude of

the second section being smaller than the first section so as to gain the advantage of improving detection ability [Col.19, Par.5, Matsugatani] . Also, since it has been held that if a technique has been used to improve one device, and a person of ordinary skill in the art would recognize that it would improve similar devices in the same way, using the technique is obvious unless its actual application is beyond his or her skill (MPEP 2143).

Regarding claim 16, LAGHEZZA as modified by Matsugatani discloses all the limitations of claim 12. LAGHEZZA does not explicitly disclose wherein, a maximum value of the maximum amplitude parameter in a first section is greater than a maximum value of the maximum amplitude parameter in a second section. Matsugatani teaches in the same field of endeavor of carrier signal modulation. Matsugatani discloses wherein, a waveform of the second section is defined by a sin-squared function, a linear function, or a rectangular function **("it is also preferable that the modulation signal oscillator 54 produces a rectangular wave including many harmonics having frequencies n.times.Fp (n.gtoreq.2)" [Col.30, ll.9-12])**

Matsugatani teaches in the same field of endeavor of carrier signal modulation. It would have been obvious to one of ordinary skill in the art prior to the effective filing date of the claimed invention to modify **LAGHEZZA** with the teachings of **Matsugatani** to incorporate the features of a rectangular function so as to gain the advantage of improving detection ability [Col.19, Par.5, Matsugatani] . Also, since it has been held that if a technique has been used to improve one device, and a person of ordinary skill in the art would recognize that it would improve similar devices in the same way, using the technique is obvious unless its actual application is beyond his or her skill (MPEP 2143).

Regarding claim 17, LAGHEZZA as modified by Matsugatani discloses all the limitations of claim 12. LAGHEZZA does not explicitly disclose wherein an amplitude is the same during the first section at a

time domain. Matsugatani teaches in the same field of endeavor of carrier signal modulation. Matsugatani discloses wherein, an amplitude is the same during the first section at a time domain **("it is also preferable that the modulation signal oscillator 54 produces a rectangular wave including many harmonics having frequencies n.times.Fp (n.gtoreq.2)" [Col.30, ll.9-12])**

Matsugatani teaches in the same field of endeavor of carrier signal modulation. It would have been obvious to one of ordinary skill in the art prior to the effective filing date of the claimed invention to modify **LAGHEZZA** with the teachings of **Matsugatani** to incorporate the features of an amplitude being the same during the first section at a time domain so as to gain the advantage of improving detection ability [Col.19, Par.5, Matsugatani] . Also, since it has been held that if a technique has been used to improve one device, and a person of ordinary skill in the art would recognize that it would improve similar devices in the same way, using the technique is obvious unless its actual application is beyond his or her skill (MPEP 2143).

Regarding claim 18, LAGHEZZA as modified by Matsugatani discloses all the limitations of claim 12. LAGHEZZA discloses, synchronizing the second section with the turn-around section between the frequency raising section and the frequency decreasing section **("The transmitting-side circuit 110 and/or the receiving-side circuit 120 may be synchronized using the timing block 130 ." [0050]).**

Regarding claim 19, LAGHEZZA as modified by Matsugatani discloses all the limitations of claim 12. LAGHEZZA discloses, generating an initial shaping signal **(" the PSK input can be provided by the reduced angle PSK (Mr-PSK) module 113." [0049])**; and converting the initial shaping signal from a digital form to an analog form of the shaping signal **("the transmit-side circuitry includes a digital signal processing circuit for processing a digitized version of the communication data and a mixer for performing frequency up-conversion." [0042]).**

5. Claims 2, 3, 13, and 14 are rejected under 35 U.S.C. 103 as being unpatentable over **LAGHEZZA(CN111880169A)** as modified by **Matsugatani(US6040796A)** as applied to claims 1, 11, and 12 above, and further in view of **ELFERINK(EP2597484A1).**

Regarding claim 2, LAGHEZZA as modified by Matsugatani discloses all the limitations of claim 1. LAGHEZZA discloses, radar apparatus wherein, the turn-around section corresponds to a junction between the frequency raising section and the frequency decreasing section **("Rectangular pulse shaping can be used so that [the communication signal] is constant for the duration" [0050] The examiner notes that the claimed turn around section is implicit to a periodic rectangular wave.)**

LAGHEZZA as modified by Matsugatani discloses a sweep carrier signal but does not explicitly disclose nor limit wherein the wherein the carrier signal is a triangle wave. ELFERINK teaches in the same field of endeavor of carrier signal modulation. ELFERINK discloses the radar apparatus wherein, the carrier signal is a triangle wave **("The radar pulse may, for example, be defined by [...] a triangular frequency-to-time relationship (i.e. one linear up-chirp followed by a corresponding linear down-chirp)" [0009])**

ELFERINK teaches in the same field of endeavor of carrier signal modulation. It would have been obvious to one of ordinary skill in the art prior to the effective filing date of the claimed invention to modify **LAGHEZZA** with the teachings of **ELFERINK** to incorporate the features of a carrier signal being a sawtooth wave so as to gain the advantage of improving range [0021, ELFERINK] . Also, since it has been held that if a technique has been used to improve one device, and a person of ordinary skill in the art would recognize that it would improve similar devices in the same way, using the technique is obvious unless its actual application is beyond his or her skill (MPEP 2143).

Regarding claim 3, LAGHEZZA as modified by Matsugatani discloses all the limitations of claim 1. LAGHEZZA discloses, radar apparatus wherein, and the turn-around section corresponds to an interval

from the frequency raising section to the frequency decreasing section **("Rectangular pulse shaping can be used so that [the communication signal] is constant for the duration" [0050] The examiner notes that the claimed turn around section is implicit to a periodic rectangular wave. There is necessarily an interval between the frequency raising section to the frequency decreasing section).**

LAGHEZZA as modified by Matsugatani discloses a sweep carrier signal but does not explicitly disclose nor limit wherein the wherein the carrier signal is a sawtooth wave. ELFERINK teaches in the same field of endeavor of carrier signal modulation. ELFERINK discloses the radar apparatus wherein, the carrier signal is a sawtooth wave **("The radar pulse may, for example, be defined by a waveform with a sawtooth-like frequency-to-time relationship (i.e. one linear up- or down chirp) [0009])**

ELFERINK teaches in the same field of endeavor of carrier signal modulation. It would have been obvious to one of ordinary skill in the art prior to the effective filing date of the claimed invention to modify **LAGHEZZA** with the teachings of **ELFERINK** to incorporate the features of a carrier signal being a sawtooth wave so as to gain the advantage of improving range [0021, ELFERINK] . Also, since it has been held that if a technique has been used to improve one device, and a person of ordinary skill in the art would recognize that it would improve similar devices in the same way, using the technique is obvious unless its actual application is beyond his or her skill (MPEP 2143).

Regarding claim 13, LAGHEZZA as modified by Matsugatani discloses all the limitations of claim 12. LAGHEZZA discloses, the signal processing method wherein, the turn-around section corresponds to a junction between the frequency raising section and the frequency decreasing section.

LAGHEZZA discloses a sweep carrier signal but does not explicitly disclose nor limit wherein the wherein the carrier signal is a triangle wave. ELFERINK teaches in the same field of endeavor of carrier signal modulation. ELFERINK discloses the signal processing method wherein, the carrier signal

is a triangle wave **("The radar pulse may, for example, be defined by [...] a triangular frequency-to-time relationship (i.e. one linear up-chirp followed by a corresponding linear down-chirp)" [0009])**

ELFERINK teaches in the same field of endeavor of carrier signal modulation. It would have been obvious to one of ordinary skill in the art prior to the effective filing date of the claimed invention to modify **LAGHEZZA** as modified by **Matsugatani** with the teachings of **ELFERINK** to incorporate the features of a carrier signal being a sawtooth wave so as to gain the advantage of improving range [0021, ELFERINK] . Also, since it has been held that if a technique has been used to improve one device, and a person of ordinary skill in the art would recognize that it would improve similar devices in the same way, using the technique is obvious unless its actual application is beyond his or her skill (MPEP 2143).

Regarding claim 14, LAGHEZZA as modified by Matsugatani discloses all the limitations of claim 12. LAGHEZZA discloses, the signal processing method wherein, and the turn-around section corresponds to an interval from the frequency raising section to the frequency decreasing section **("Rectangular pulse shaping can be used so that [the communication signal] is constant for the duration" [0050] The examiner notes that the claimed turn around section is implicit to a periodic rectangular wave. There is necessarily an interval between the frequency raising section to the frequency decreasing section).**

LAGHEZZA discloses a sweep carrier signal but does not explicitly disclose nor limit wherein the wherein the carrier signal is a sawtooth wave. ELFERINK teaches in the same field of endeavor of carrier signal modulation. ELFERINK discloses the radar apparatus wherein, the carrier signal is a sawtooth wave **("The radar pulse may, for example, be defined by a waveform with a sawtooth-like frequency-to-time relationship (i.e. one linear up- or down chirp) [0009])**

ELFERINK teaches in the same field of endeavor of carrier signal modulation. It would have been obvious to one of ordinary skill in the art prior to the effective filing date of the claimed invention to modify **LAGHEZZA** as modified by **Matsugatani** with the teachings of **ELFERINK** to incorporate the

features of a carrier signal being a sawtooth wave so as to gain the advantage of improving range [0021, ELFERINK] . Also, since it has been held that if a technique has been used to improve one device, and a person of ordinary skill in the art would recognize that it would improve similar devices in the same way, using the technique is obvious unless its actual application is beyond his or her skill (MPEP 2143).

6. **Claim 10** is rejected under 35 U.S.C. 103 as being unpatentable over **LAGHEZZA(CN111880169A)** as modified by **Matsugatani(US6040796A)** as applied to claims 1, 11, and 12 above, and further in view of **ELFERINK(EP2597484A1) ADIB(US20170074980A1)**

Regarding claim 10, LAGHEZZA as modified by Matsugatani discloses all the limitations of claim 1. LAGHEZZA does not explicitly disclose nor limit wherein the wherein the radar apparatus is an ultra wideband (UWB) radar. ADIB teaches in the same field of endeavor of carrier signal modulation. ADIB discloses the, radar apparatus wherein, the radar apparatus is an ultra wideband (UWB) radar **("the signal generator is an ultra wide band frequency modulated carrier wave (FMCW) generator" [0079])**

ADIB teaches in the same field of endeavor of carrier signal modulation. It would have been obvious to one of ordinary skill in the art prior to the effective filing date of the claimed invention to modify **LAGHEZZA** as modified by **Matsugatani** with the teachings of **ADIB** to incorporate the features of an ultra wideband (UWB) radar so as to gain the advantage of reducing the effects of noise [0091, ADIB] . Also, since it has been held that if a technique has been used to improve one device, and a person of ordinary skill in the art would recognize that it would improve similar devices in the same way, using the technique is obvious unless its actual application is beyond his or her skill (MPEP 2143).

7. For applicant's benefit portions of the cited reference(s) have been cited to aid in the review of the rejection(s). While every attempt has been made to be thorough and consistent within the rejection

it is noted that the *PRIOR ART MUST BE CONSIDERED IN ITS ENTIRETY, INCLUDING DISCLOSURES THAT TEACH AWAY FROM THE CLAIMS.* See MPEP 2141.02 VI.

Conclusion

8. Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any nonprovisional extension fee (37 CFR 1.17(a)) pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of this final action.

9. Any inquiry concerning this communication or earlier communications from the examiner should be directed to CLAYTON PAUL RIDDER whose telephone number is (571)272-2771. The examiner can normally be reached Monday thru Friday ET.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Jack Keith can be reached on (571) 272-6878. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/C.P.R./
Examiner, Art Unit 3646

/JACK W KEITH/
Supervisory Patent Examiner, Art Unit 3646